ECE 340
Test #1 Review

Problem #1

A silicon sample with $N_D = 10^{16}$ cm^{-3} donors is optically excited such that 10^{19} EHP/cm^3 are generated per second uniformly in the sample. The laser causes the sample to heat up to 450 K. The electron and hole lifetimes are both 10^{-6} seconds, $D_p = 12$ cm^2/s; $D_n = 36$ cm^2/s; and $n_i = 10^{14}$ cm^{-3} at T = 450 K. Find the quasi-Fermi levels and the change in the conductivity of the sample upon shining the light.

Problem #2

A portion ($0 \leq x \leq L$) of a silicon sample, uniformly doped with $N_D = 10^{15}$ cm^{-3} donors and maintained at room temperature, is subject to a steady state perturbation such that:

$$n \approx N_D$$
$$p = n_i(1 - x/L) + n_i^2/N_D$$

Since $n \approx N_D$, it is reasonable to assume that the electric field is approximately equal to zero in the region of interest. Given this, sketch the energy band diagram for the perturbed region specifically including E_c, E_i, E_v, F_n, and F_p on the diagram.

Problem #3

Given a bar-shaped layer of width W, length L, and depth d, and assuming an arbitrary $N_D(x)$ variation with the depth x from the surface of the layer, derive a formula to calculate the resistance of the layer.

Problem #4

The earth is hit by a mysterious ray that momentarily wipes out all minority carriers. Majority carriers are not affected. Initially in equilibrium and not affected by room light, a uniformly doped silicon wafer sitting on your desk is struck by the ray at time t = 0. The wafer doping is $N_A = 10^{16}$ cm^{-3}, $\tau_n = 10^{-6}$ s, and T = 300 K.

A. What is δn at $t = 0^+$ (i.e. just after time t = 0)?
B. Does generation or recombination dominate at $t = 0^+$?
C. Do low-level injection conditions exist inside the wafer at $t = 0^+$?
D. Find a formula for $\delta n(t)$ for $t > 0$.

Problem # 1

Find $F_n - F_p$ and $\Delta\sigma$.

light induced EHP: $\delta n = \delta p = g_{op} \cdot \tau = \left(10^{19} \frac{1}{cm^3 \cdot s}\right)\left(10^{-6}\, s\right) = 10^{13}\, cm^{-3}$

we have L.L.I. $N_D \gg \delta n$

$$n = n_0 + \delta n = N_D + 10^{13}\, cm^{-3} \approx 10^{16}\, cm^{-3}$$

$$p = p_0 + \delta p = \frac{n_i^2}{n_0} + 10^{13}\, cm^{-3} \approx 10^{13}\, cm^{-3}$$

$k_B T$ @ 450 K : 0.039 eV

$$F_n - E_i = k_B T \ln\left(\frac{n}{n_i}\right) = 0.039\, eV \cdot \ln\left(\frac{10^{16}\, cm^{-3}}{10^{14}\, cm^{-3}}\right) = 0.18\, eV$$

$$E_i - F_p = k_B T \ln\left(\frac{p}{n_i}\right) = 0.039\, eV \cdot \ln\left(\frac{10^{13}\, cm^{-3}}{10^{14}\, cm^{-3}}\right) = -0.090\, eV$$

$$\boxed{F_n - F_p = 0.090\, eV}$$

$$\mu_n = \frac{D_n}{\left(\frac{k_B T}{q}\right)} = \frac{36\, \frac{cm^2}{s}}{0.039\, V} = 927\, \frac{cm^2}{V \cdot s}$$

$$\mu_p = \frac{D_p}{\left(\frac{k_B T}{q}\right)} = \frac{12\, \frac{cm^2}{s}}{0.039\, V} = 309\, \frac{cm^2}{V \cdot s}$$

$$\boxed{\Delta\sigma = q(\mu_n \delta n + \mu_p \delta p) = 0.00198\, \frac{1}{\Omega \cdot cm}}$$

Problem # 2

(i) $\mathcal{E}(x) = 0 \rightarrow \frac{\partial E_C}{\partial x} = \frac{\partial E_V}{\partial x} = \frac{\partial E_i}{\partial x} = 0$

(ii) $n \approx N_D \rightarrow F_N \approx E_F$

$$E_F - E_i = k_B T \ln\left(N_D/n_i\right) = (0.0259\ \text{eV}) \ln\left(\frac{10^{15}\ \text{cm}^{-3}}{10^{10}\ \text{cm}^{-3}}\right) = 0.298\ \text{eV}$$

(iii) $p = p_0 + \delta_p = \frac{n_i^2}{N_D} + n_i\left(1 - x/L\right) = n_i\, e^{\frac{E_i - F_p}{k_B T}}$

$$p_0 = \frac{n_i^2}{N_D} = 10^5\ \text{cm}^{-3} \qquad \delta_p = n_i\left(1 - x/L\right) = 10^{10}\left(1 - x/L\right)$$

@ $x = 0$: $p \approx n_i$ & $F_p = E_i$

@ $x = L$: $p = p_0$ & $F_p = E_F$

For most values : $\delta_p \gg p_0$

$$e^{\frac{E_i - F_p}{k_B T}} \approx 1 - \frac{x}{L}$$

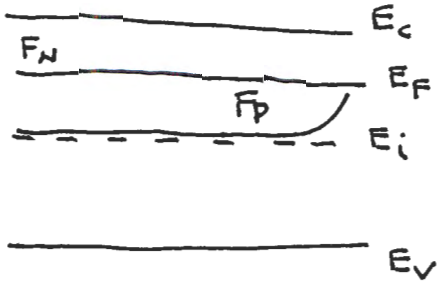
$$E_i - F_p = k_B T \ln\left(1 - \frac{x}{L}\right) \rightarrow$$

$F_p - E_i = 0.0027$ eV if $x/L = 0.1$

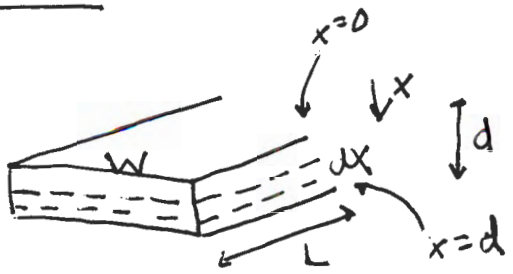
$F_p - E_i = 0.018$ eV if $x/L = 0.5$

$F_p - E_i = 0.060$ eV if $x/L = 0.9$

$F_p \approx E_i$ until $x = L$



Problem # 3



$$R = \frac{1}{G} = \frac{\rho L}{A} \qquad \text{but} \qquad N_D \rightsquigarrow \rho = \frac{1}{q_0 \mu_n N_D}$$

$$dG = d(1/R) = \frac{W}{L\rho}\,dx = \frac{W}{L} q_0 \mu_n N_D\, dx$$

Sections are in parallel so they add.

$$G = \sum dG = \int_0^d \frac{W}{L} q_0 \mu_n N_D\, dx \quad \text{or}$$

$$R = \frac{1}{G} = \frac{L/W}{q \int_0^d \mu_n N_D\, dx}$$

Problem # 4

a) $\delta n = n - n_0 \rightarrow \delta n = -n_0 \qquad n_0 = \frac{n_i^2}{N_A} = \frac{10^{20}}{10^{16}} = 10^4\ cm^{-3}$

b) Generation @ $t = 0^+$ there is a diffusion of carriers.

c) Yes $|\delta n| = 10^4\ cm^3 \ll P_0 = N_A = 10^{16}\ cm^3$

d) $\frac{\partial \delta n(t)}{\partial t} = D_N \underbrace{\frac{\partial \delta n(t)}{\partial x^2}}_{=0 \text{ because no gradient}} - \frac{\delta n(t)}{\tau_n}$

$$\frac{\partial \delta n}{\partial t} = -\frac{\delta n}{\tau_n} \qquad \text{need B.C.} \quad \delta n(0) = -n_0$$

$$\delta n(t) = A e^{-t/\tau_n} \rightarrow -n_0 e^{-t/\tau_n}$$